# MIT 8103 Advanced Database Systems

## Case study: JP Morgan Chase Nigeria Commercial Banking Operations (Nigeria) | Finacle & Polyglot Architecture

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| **Course:** | MIT 8103 — Advanced Database Systems | **Academic Session:** | 2026 / 2027 |
| **Repository:** | github.com/Cookeygam/FRANCIS_COOKEY_MIT-8103_ADVANCED_DB_ASSESSMENT | | |

**Database used for hands-on work:** Oracle Database 19c Enterprise Edition (SQL, PL/SQL). Evidence of version, sample data and test output is in the repository.

## 1. Executive Overview & Architectural Context

This executive summary synthesizes the design, implementation, optimization, and governance work completed across five practical database engineering portfolios. Modeled on JPMorgan Chase & Co.'s commercial banking operations in Nigeria under Central Bank of Nigeria (CBN) regulatory oversight, the solution mirrors an enterprise Infosys Finacle core banking environment backed by Oracle Database 19c and polyglot NoSQL extensions.

To enforce strict separation of duties and least-privilege access control, the relational core is partitioned into three isolated security schemas: **TRANSACT** (corporate clearing, multi-currency accounts, intraday/historical transaction processing), **LEDGER** (general ledger balancing, EOD closure, nostro/vostro reconciliation), and **CUSTOMER** (CBN/CAC corporate registry, KYC profiles, and immutable AML forensic audit trails), governed via three dedicated database roles ( *ROLE_CORPORATE_CLEARING* , *ROLE_FINANCIAL_CONTROL* , and *ROLE_RISK_COMPLIANCE_AUDIT* ).

## 2. Organization

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| Scenario | A simulated commercial banking operation: JPMorgan Chase & Co. opening in Nigeria under a Central Bank of Nigeria (CBN) license. JPMorgan Chase is used only to make the setting realistic. All data is synthetic and nothing here describes the bank's real systems. |
| Nature of the system | The transactional core of the bank: heavy payment traffic at peak times, large batch jobs, and no loss of committed transactions. Business rules sit in the database itself (constraints, atomic transactions) so an application fault or dropped connection cannot leave bad data behind. |
| Users | Corporate Clearing team (payments), Financial Control team (ledger, reconciliation), Risk, Compliance and Audit team (CBN, AML/KYC). Each has its own schema or role. |
| Major data stored | Retail and corporate customers, branches, accounts and balances, a double-entry daily and history transaction journal, internal ledger accounts (asset, liability, nostro), and an AML forensic trail. |
| Key operations | Post a transfer so debit, credit and audit succeed or fail together; report high-value accounts with high AML risk; balance the ledger; investigate suspicious activity; keep an append-only audit trail. |

| Criterion | Repository Artifact | Implemented Engineering |
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| Relational Data Modeling & Integrity | portfolio-1-database-design-and-modelling/ ( schema_DB_design.sql , normalization.md, constraint_test.sql ) | Normalized 3NF banking schema across TRANSACT, LEDGER, and CUSTOMER; enforced PK/FK integrity, NUBAN uniqueness, check constraints, and ERD documentation. |
| Query Processing & Indexing | portfolio-2-query-optimisation/ (portfolio_2_queries.sql, query_optimization_sql.md) | Multi-table AML compliance query combining DAILY_TXN_DTL and HISTORY_TXN_DTL; deployed 5 B-Tree indexes with key compression and refreshed DBMS_STATS. |
| Transactions & Concurrency Control | portfolio-3-transactions-concurrency/ ( TRANSACTION_concurrency.md , portfolio_3_overview.md) | ACID-compliant PL/SQL engine using READ COMMITTED isolation, pessimistic row locking “UPDATE NOWAIT", lien verification, and atomic exception rollbacks. |
| NoSQL & Advanced Data Models | portfolio-4-nosql/ ( portfolio_4_mongodb.js , transaction_document_model.json ) | MongoDB document model for high-velocity nested transaction, flexible JSON payloads, schema validation rules. |
| Distributed & Cloud Architecture | portfolio-5-distributed-cloud/ (Distributed_Banking_Database_Architecture.md) | High-availability design balancing CAP trade-offs, synchronous vs. asynchronous replication for zero RPO on financial ledgers. |

## 3. Technicalities Encountered & Engineering Solutions

**Portfolio 1**: Relational Schema Design & Cross-Domain Governance — 3NF / RBAC
Challenge: Model multi-currency corporate accounts, intraday and archived ledger entries, and compliance audit logs without update anomalies or privilege leaks.
**Solution**: Normalized entities to 3NF across three schema boundaries. Enforced domain integrity using CHECK constraints for currency codes, soft-delete flags (del_flg), and risk-score bounds; validated invalid inserts in constraint_test.sql.

**Portfolio 2**: AML Query Tuning & Covering Index Design — Oracle 19c
Challenge: Flagging corporate NGN accounts with balances ≥ ₦250M and AML risk scores ≥ 7.5 caused duplicate balances from append-only AML_FORENSIC_TRAIL joins, plus full table scans and PGA memory sorts on HISTORY_TXN_DTL.
**Solution**: Isolated the latest AML event per customer using window functions. Unified intraday and archived 30-day transactions with UNION ALL, then built five covering B-Tree indexes with COMPRESS 1/2 and cut median latency from 2,140 ms to 72 ms.

**Portfolio 3**: High-Concurrency Settlement & Pessimistic Locking — ACID - PL/SQL
Challenge: Concurrent debit requests during peak RTGS settlement and EOD closure risked lost updates, race conditions on lien-encumbered balances, and session hangs.
**Solution**: Built an atomic PL/SQL procedure under READ COMMITTED using SELECT ... FOR UPDATE NOWAIT for row-level locking on GENERAL_ACCT_MAST_TABLE. Exception handlers managed ORA-00054 lock contention and frozen profiles, executing immediate ROLLBACK to prevent partial writes to DAILY_TXN_DTL or AML_FORENSIC_TRAIL.

**Portfolio 4**: NoSQL Modeling for Unstructured Data— MongoDB
Challenge: Rigid relational schemas could not efficiently ingest polymorphic payment payloads—SWIFT, NIP, RTGS metadata, device fingerprints, and nested AML triggers—without frequent ALTER TABLE migrations.
**Solution**: Designed a document-oriented MongoDB model embedding audit arrays and channel-specific attributes in atomic documents. Enforced validation with Json Schema and used compound indexes for fast compliance queries.

**Portfolio 5**: Distributed Resilience, Replication & Sharding — CAP
Challenge: Scale banking workloads across availability zones while meeting CBN data-residency rules and preventing split-brain ledger inconsistencies during partitions.
**Solution**: Designed a hybrid topology prioritizing CP (Consistency + Partition Tolerance) for core balance mutations using synchronous Oracle Data Guard Maximum Availability replication. Distributed high-volume transaction history using hashed/range sharding keys (cif_id / acct_id) to prevent hot spotting.

## 4. References

Oracle Database 19c documentation (SQL Language Reference, Concepts, Performance Tuning Guide). Oracle: schema design, docs.oracle.com. MongoDB: data modelling, replication, sharding and aggregation manuals, mongodb.com/docs. Mermaid: entity relationship diagrams, mermaid.js.org. MIT 8103 Continuous Assessment Brief, Miva Open University, 2026/2027.